

Effect of the plastic pollutant bisphenol A on the biology of aquatic organisms: A meta-analysis

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Abstract

Plastic pollution is a global environmental concern. In particular, the endocrine-disrupting chemical bisphenol A (BPA) is nearly ubiquitous in aquatic environments globally, and it continues to be produced and released into the environment in large quantities. BPA disrupts hormone signalling and can thereby have far-reaching physiological and ecological consequences. However, it is not clear whether BPA has consistent effects across biological traits and phylogenetic groups. Hence, the aim of this study was to establish the current state of knowledge of the effect of BPA in aquatic organisms. We show that overall BPA exposure affected aquatic organisms negatively. It increased abnormalities, altered behaviour and had negative effects on the cardiovascular system, development, growth and survival. Early life stages were the most sensitive to BPA exposure in invertebrates and vertebrates, and invertebrates and amphibians seem to be particularly affected. These data provide a context for management efforts in the face of increasing plastic pollution. However, data availability is highly biased with respect to taxonomic groups and traits studies, and in the geographical distribution of sample collection. The latter is the case for both measurements of the biological responses and assessing pollution levels in water ways. Future research effort should be directed towards biological systems, such as studying endocrine disruption directly, and geographical areas (particularly in Africa and Asia) which we identify to be currently undersampled.

KEYWORDS

BPA, dose-response, ecotoxicology, endocrine-disrupting chemical (EDC), reproduction, survival

1 | INTRODUCTION

The large-scale production of plastic is of growing concern for ecosystem function (Thompson, Moore, Saal, & Swan, 2009). In particular, the growing accumulation of global plastic waste in aquatic environments has major impacts on wildlife (Lebreton et al., 2017; Ostle et al., 2019). Large plastic pieces can entangle wildlife or are ingested, which causes mortality (Derraik, 2002; Santos, Andrades, Boldrini, & Martins, 2015; Votier, Archibald, Morgan, & Morgan, 2011). Microplastics (<5 mm size) that break down from larger plastics are ingested by small organisms and

are accumulated through the food chain (Foley, Feiner, Malinich, & Höök, 2018; Galloway, Cole, & Lewis, 2017; Nelms, Galloway, Godley, Jarvis, & Lindeque, 2018; Wright, Thompson, & Galloway, 2013). Importantly, and more insidiously, chemical leachates from plastics can interfere with a myriad of biological functions (Oehlmann et al., 2009). Plastic leachates have been shown to contain carcinogens (Rowdhwil & Chen, 2018) and endocrine-disrupting chemicals (EDCs) that alter hormone function (Colborn, Saal, & Soto, 1993; Diamanti-Kandarakis et al., 2009; Roy, Chakraborty, & Chakraborty, 2009). The most common EDC is the chemical plasticizer bisphenol A (BPA), which is used in the

production of epoxy resins and polycarbonate plastics (Corrales et al., 2015; Staples, Dorn, Klecka, O'Block, & Harris, 1998).

BPA enters into freshwater and marine systems by leaching from BPA-based products, and also from effluent discharges in manufacturing plants, wastewater treatment plants and land-fill sites (Corrales et al., 2015; Huang et al., 2012; Kang, Aasi, & Katayama, 2007). BPA has a low potential for accumulation in freshwater environments and degrades relatively quickly with a half-life of ~4.6 days under aerobic conditions (Cousins, Staples, Klečka, & Mackay, 2002; Im & Löffler, 2016; Staples et al., 1998). In seawater, BPA does not seem to degrade readily (>30 days), suggesting that BPA has a higher accumulation potential in marine environments (Danzl, Sei, Soda, Ike, & Fujita, 2009; Kang & Kondo, 2005). However, even in freshwater environments, the persistence of plastic waste and the continuous effluent discharges into the environment means that aquatic organisms are consistently exposed to BPA (Sharma et al., 2009). BPA is taken up by aquatic organisms via the gills, ingestion of food and absorption in the digestive tract, cutaneous uptake or plant root absorption (Bjerregaard, Andersen, Pedersen, Pedersen, & Korsgaard, 2007; Chen et al., 2017; Guo et al., 2017; Völkel, Colnot, Csanády, Filser, & Dekant, 2002).

BPA can affect numerous biological functions. Like other EDCs, BPA disrupts signalling pathways, including (but not limited to) those mediated by thyroid hormone receptors, and oestrogen-related receptors (Rogers, Metz, & Yong, 2013). The most common pathological conditions include feminization via induced vitellogenin (vtg) production in males (Van den Belt, Verheyen, & Witters, 2003), increased abnormalities of reproductive systems (Segner, Navas, Schäfers, & Wenzel, 2003; Wang, Yang, et al., 2019) and deformities in developing embryos (Warner & Jenkins, 2007). Notably, BPA can interfere with thyroid hormone signalling pathways (Rubin, 2011), thereby disrupting energy metabolism and muscle function. The consequence of BPA exposure correlates with reductions in behavioural activities (Yokota et al., 2000), cardiovascular function (Schirling, Bohlen, Tribskorn, & Köhler, 2006), development and growth (Aluru, Leatherland, & Vijayan, 2010) and locomotor performance (Little & Seebacher, 2015). BPA exposure during development can also have persistent effects into adulthood via epigenetic modifications (Jašarević et al., 2011), suggesting that short-term exposures to BPA have lasting consequences on life history (Sadoul, Augustine, Zimmer, Begout, & Vijayan, 2019; Thomas, Birceanu, Sadoul, & Vijayan, 2018).

Even though the potential impacts of BPA are known, the generality of these impacts across phylogenetic groups and biological systems is unresolved. A quantitative assessment of known impacts of BPA will be invaluable to inform wildlife management and conservation decisions. For example, the success of conservation measures of species sensitive to BPA in polluted areas will be much reduced. Hence, more effective management decisions can be implemented if species sensitivities and specific impacts of BPA are known. To facilitate this process, we conducted a meta-analysis of existing literature to quantify the effects of BPA on aquatic organisms. A meta-analytic approach is important particularly

because the effects of BPA on biological functions can be context dependent, and different taxa and biological traits show different responses when exposed to BPA (Abdel-Tawwab & Hamed, 2018; Martínez et al., 2019; Staples, Woodburn, Caspers, Hall, & Klecka, 2002; Wright-Walters, Volz, Talbott, & Davis, 2011). Given the ongoing development of policies regarding the environmental consequences of EDCs (Canesi & Fabbri, 2015; Crain et al., 2007; Flint, Markle, Thompson, & Wallace, 2012), there is an urgent need to evaluate environmental concentrations and their impact on aquatic organisms.

We aimed to provide a quantitative synthesis of the current state of knowledge on global environmental BPA levels and their effects on aquatic organisms. Specifically, we aimed to (a) summarize global BPA levels in both natural surface waters and aquatic organisms, (b) investigate how BPA exposure affects different functional traits across different levels of biological organization (cellular to organism level response) and (c) examine taxon- and development-specific responses to BPA exposure. Our meta-analysis identified important gaps in knowledge and we suggest future research directions.

2 | METHODS

2.1 | Data collection and effect size calculation

We followed PRISMA guidelines (Moher et al., 2015) in the design and analysis of this study. We searched peer-reviewed studies in the Web of Science, ScienceDirect and Scopus on 10 July 2019. The search terms are given in Figure S1. Studies were included when they: (a) reported means, sample size (n) and variation as either standard deviation (SD), standard error (SE) or confidence intervals (95% CI) for both the control group (no BPA present, vehicle/solvent control) and experimental group/s (BPA-exposed); (b) stated the relevant BPA concentrations (with appropriate units) and exposure period and (c) contained at least one response variable that fitted our defined functional trait groups (see Section 2.2 for more detail). Detailed description of the data descriptors (Table S1) and exclusion criteria are presented in Data S1. Forward (papers identified citing the original paper) and backward (previous papers identified that the original paper cited) searches were obtained from Google Scholar until 4 October 2019. The final data set comprised a total of 3,141 individual effects sizes from 179 studies across 93 species (Figure S1). Data from text and tables were directly extracted from the study, while data from figures were digitized via the *metaDigitise* package in R (Pick, Nakagawa, & Noble, 2019). For studies presenting SE only, SD was calculated as $SD = SE \times \sqrt{n}$, and if 95% CI were presented, we calculated the SD as $SD = \sqrt{n} \times ((\text{upper } 95\% \text{ CI} - \text{lower } 95\% \text{ CI}) / 3.92)$. For tank-replicated experiments, sample size was obtained at the tank level to negate the non-independence effects of tank treatments.

Effect sizes (k) were calculated as the natural log-transformed response ratio ($\ln RRR$; Hedges, Gurevitch, & Curtis, 1999) and the sampling variance (v) was calculated following Nakagawa, Noble, Senior, and Lagisz (2017):

$$\ln RR = \ln \left(\frac{\bar{x}_T}{\bar{x}_C} \right),$$

$$v = \frac{(SD_T)^2}{N_T \bar{x}_T^2} + \frac{(SD_C)^2}{N_C \bar{x}_C^2},$$

where $\bar{x}_T$, SD_T and N_T represent mean responses, SD and sample size for the BPA treatment group, while $\bar{x}_C$, SD_C and N_C , represent mean responses, SD and sample size for the control groups. We added +1 to all mean responses to allow for calculations with values of zeros (see Table S1). To account for low-sample size biases, we applied Lajeunesse (2015) Delta method for studies with sample size of less than three:

$$\ln RR^A = RR + \frac{1}{2} \left[\frac{(SD_T)^2}{N_T \bar{x}_T^2} + \frac{(SD_C)^2}{N_C \bar{x}_C^2} \right],$$

$$v^A = v + \frac{1}{2} \left[\frac{(SD_T)^4}{N_T^2 \bar{x}_T^4} + \frac{(SD_C)^4}{N_C^2 \bar{x}_C^4} \right].$$

2.2 | Moderator variables

Moderator (explanatory) variables were included to test for correlations with BPA-induced responses. The moderator variables included BPA concentration (corrected to $\mu\text{g/L}$), exposure period (corrected to days), population source (captive or wild), exposure temperature (as degree Celsius; $^{\circ}\text{C}$), developmental stage of exposure (egg, larvae, juvenile, adult or unknown; Table S2) and sex of the treatment groups (male, female, both, unknown or not applicable). Biological responses to BPA exposure were grouped in the following categories: (a) biological levels were divided into five categories: organism, organ, tissue, cellular and macromolecule level responses (see Table S3 for descriptors); (b) traits were grouped into 12 broad categories: abnormality and damage, behaviour, cardiovascular, development, energy metabolism, growth, immune function, locomotion, plant-specific, survival, reproduction and thyroid response (see Table S4 for descriptors) and (c) nine taxonomic groups were defined: amphibians, angiosperms, crustaceans, fish, insects, macroalgae, molluscs, other invertebrates, phytoplankton. 'Other invertebrates' (annelids, platyhelminthes, echinoderms, ascidians, porifera, cnidaria, rotifers) were combined due to low sample sizes in the literature. All specific responses for each trait group are presented and described in Table S4. We only included specific responses if the effect size sample was >5 (k) to be included in analyses.

2.3 | BPA levels in the environment and within organisms

We also extracted BPA levels in the surface water of freshwater, estuary and marine environments, and in aquatic organisms from the literature. To assess the geographic distribution of known BPA

levels, we plotted results separately for each country where data were collected (Figure S1). When possible, the mean (or geometric mean or median), minimum (or lower 95% CI) and maximum (or upper 95% CI) BPA levels from each site or organism per study were collected. All BPA levels were converted to the same unit of ng/L or ng/g for the environment and organisms, respectively. Studies that included BPA levels below the detection limit were given values at their detection limit (see Data S1 for handling of data below detection limits), and non-detected samples were treated as 0 ng/L or 0 ng/g . BPA concentrations from anthropogenic sites such as waste/landfills, wastewater, treatment plants, drinking water, effluent discharge points and water catchment stations were excluded due to high disturbances in the environment as well as low biodiversity in those areas. Additionally, summary BPA levels presented in government reports (e.g. Staples et al., 2018) that were not published were not included. To summarize BPA levels in aquatic organisms, we only included studies that reported the exact sample collection site in the natural habitat of the organism. Organisms were grouped into taxonomic groups (crustaceans, fishes, mammals, molluscs, other invertebrates, plankton, reptiles). 'Other invertebrates' (annelid, insects, flatworms) were grouped together due to low sample size.

As BPA is known to leach into the surrounding environment from plastic materials (Cooper, Kendig, & Belcher, 2011), we collected data on the total plastic generated per country, and the percentage of inadequately managed plastic waste per country (defined as not formally managed and includes disposal in dumps or open, uncontrolled landfills, where it is not fully contained), which have high risks of polluting rivers, lakes and oceans. The total plastic waste generated per country (t/year) and inadequately managed plastic waste (%) data from each country for the year 2010 were obtained from Jambeck et al. (2015). A linear model (lm) from R default stats package was used to examine the relationship between BPA concentrations in natural surface waters with either the total plastic waste generated per country or inadequately managed plastic waste per country. Additionally, the mean plastic input from rivers into coastal regions presented as metric tonnes per year (t/year) was obtained from Lebreton et al. (2017).

2.4 | Statistical analysis

All analyses and figure production were performed in R 3.5.1 (R Core Team, 2018). Meta-analysis data were presented as mean effect size $\pm$ 95% CI, while BPA levels in the environment and in aquatic organisms were presented as the geometric mean $\pm$ SD to account for high spatial variation in BPA levels across aquatic regions and organisms. Sample sizes in Section 3 refer to individual effect sizes.

2.4.1 | Post-data collection processing

For the meta-analysis of means, effect sizes were coded so that negative values indicate a detrimental effect of BPA. For example,

survival and mortality represent the same response in opposite directions so that mortality was therefore converted to negative values indicating that it affects the organism negatively (Table S4). Additionally, effect sizes with sample size of one (no variation) were excluded post-data collection. To correct for phylogenetic non-independence, we built a phylogeny using the Open Tree Taxonomy nomenclature format and retrieved the phylogenetic relationships from the Open Tree of Life (Hinchliff et al., 2015) using the *rotl* package (Michonneau, Brown, & Winter, 2016). Polytoimy was accounted for via randomization using the function 'multi2di', and branch lengths were estimated using the function 'compute.brlen' from the *ape* package (Paradis & Schliep, 2018). The generated ultrametric tree was converted to a phylogenetic relatedness correlation matrix for subsequent analysis (Figure S3).

2.4.2 | Meta-analyses and meta-regression

We first constructed a multi-level meta-analytic model (only with the intercept) with four random factors to assess non-independence in the data (Nakagawa et al., 2017): (a) species identity, to account for similarities of effect sizes within the same species; (b) phylogeny, to account for similarity due to common ancestors; (c) study ID, to account for multiple effects per study; and (d) observation (effect size) level random effect, which is equivalent to the residual term in a normal lm. For phylogenetic non-independence, we ran both a non-phylogenetic multi-level (three random factors as stated above) model and phylogenetic multilevel (four random factors as stated above) model using the 'rma.mv' function from the *metafor* package (Viechtbauer, 2010). For both multi-level models, we calculated an extended heterogeneity (I^2) statistic following Nakagawa and Santos (2012) to partition total heterogeneity (I^2 total) into within-species (I^2 species) phylogenetic variance (I^2 phylogeny; for the phylogenetic multilevel model only), study ID variance (I^2 study) and residual variance (I^2 residual). As both models had equally high heterogeneity (0.99), and no difference in AIC values and the phylogenetic signal was low ($\lambda < 0.0001$, $p = 1$), we proceeded with non-phylogenetic meta-regression models. We then created a set of meta-regression models for five specific biological moderators, all of which had species, study ID and residual variance as random effects. The five categorical moderators were biological level, functional traits, taxonomic groups, developmental stage and sex. Three moderators (biological level, functional traits and developmental stage) showed significant effects, which we further analysed in subsets. For the biological level moderator, we examined differences in effect size for each functional trait across different biological levels. For the functional trait moderator, we examined specific responses within each functional trait category. The top five specific responses with the most effect size observations (k) were subset for each functional trait category and analysed. For the development stage moderator, we examined differences in effect size for each developmental period across taxonomic groups to observe taxon-specific differences in

developmental response to BPA. For continuous moderators, we included BPA dosage, exposure period and test temperature in the regression model. For all meta-regression models, we calculated R^2 as the marginal R^2 in mixed models described in Nakagawa and Schielzeth (2013). In meta-analysis, R^2 represents of variance accounted for after taking away sampling error.

2.4.3 | Publication bias and sensitivity analysis

We assessed publication bias at the mean level using three different methods. First, we fitted a meta-regression model with year of publication as a moderator. Evidence of time-lag bias is indicated if the 95% CI of the slope estimate excludes zero. Second, we used funnel plots to visualize possible publication bias (evident by funnel plot asymmetry), by plotting the effect sizes and the meta-analytic residuals against their precision. Third, we performed Egger's regression (Egger, Smith, Schneider, & Minder, 1997) on the meta-analytic residuals obtained from the overall model to formally test for evidence of funnel plot asymmetry, by fitting an lm of the meta-analytic residuals against their precision. If the intercept of the Egger's regression is significantly different from 0 (at $\alpha = 0.05$), publication bias may be present. We then performed a trim-and-fill analysis using the 'trimfill' function in *metafor* to estimate the number of effect sizes potentially missing from our data set. Asymmetry can also be due to heterogeneity in the data, and thus, any correction would erroneously bias the true mean effect size (Peters, Sutton, Jones, Abrams, & Rushton, 2007). Therefore, we used meta-analytic residuals described above to determine the presence of publication bias instead of weighted effect sizes as suggested by Nakagawa and Santos (2012). For the sensitivity analysis, we used Cook's distance via the 'cooks.distance' function in *metafor* to estimate the influence of data points on our overall model. Publication bias and Cook's distance results are presented in Data S1 and in Figure S4a–d. Overall, our analyses revealed no strong evidence that our results are influenced by publication bias.

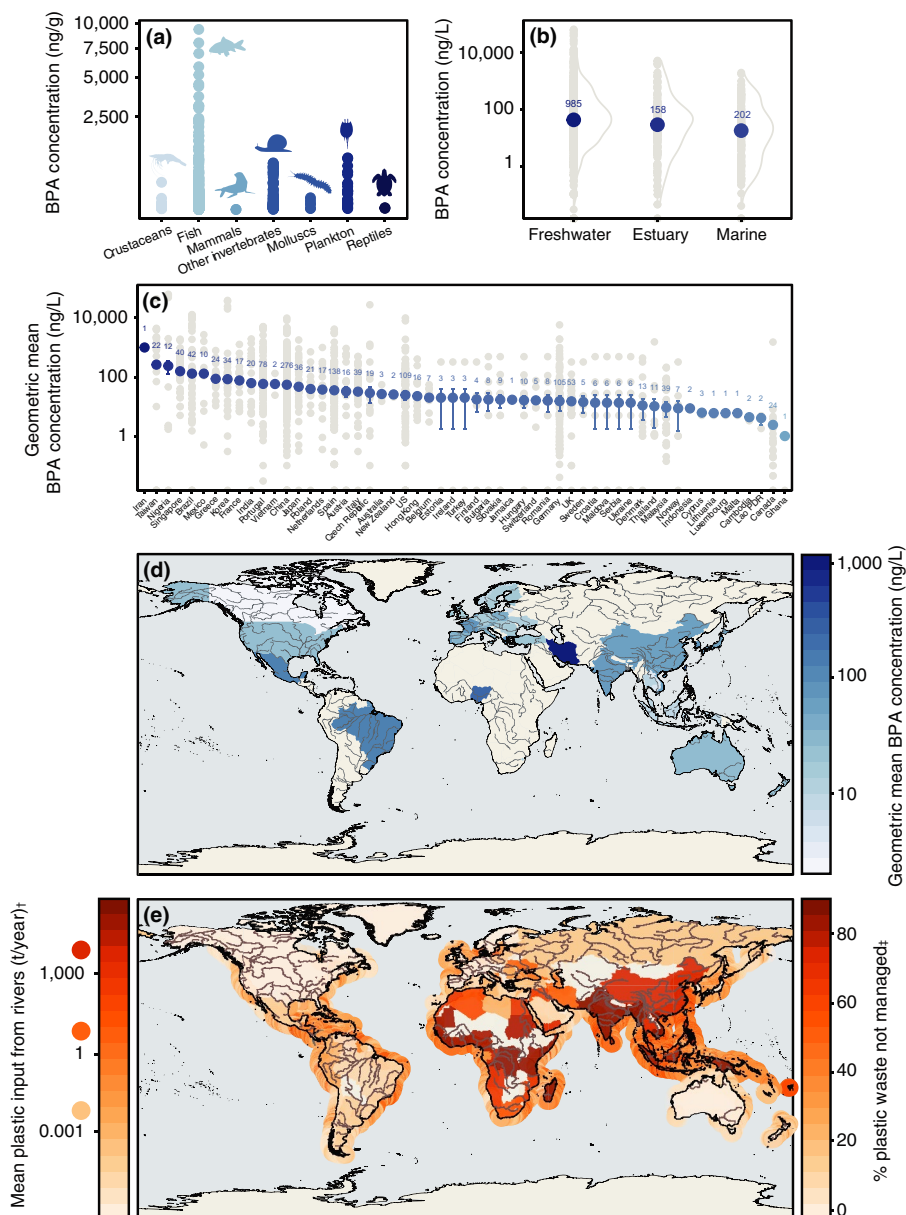
3 | RESULTS

3.1 | BPA levels in aquatic organisms and the environment

BPA levels in aquatic organisms were observed in seven taxonomic groups (162 species) across 31 countries, ranging from not detected to 9,340 ng/g (geometric mean = 13.97 ± 9.43 ng/g). Most of the BPA levels were measured in fish, which made up 71% (117 species) of all the species measured (Table S5). Other taxonomic groups were poorly represented in the literature (Figure 1a).

There were 55 countries for which surface water BPA levels were published, and levels ranged from not detected to 63,640 ng/L (geometric mean = 35.5 ± 9 ng/L). Freshwater ecosystems had the highest average BPA levels compared to estuaries, while marine/coastal

FIGURE 1 Bisphenol A (BPA) concentrations in natural surface waters and in aquatic organisms in situ in their natural habitat, and spatial distribution of plastic waste. (a) BPA concentration (ng/g) in various aquatic organisms. Data are presented as individual data points, and silhouettes represent example organisms in each taxon. Note that there were no data of concentrations in wild animals in the literature for some taxa (e.g. amphibians). (b) Geometric mean ($\pm$ SD and sample size) of BPA concentration (ng/L) in natural surface waters of either freshwater, estuary and marine environments. Individual BPA levels presented as grey points with the distribution of the BPA levels shown in the grey violin plot. (c, d) Geometric mean ($\pm$ SD and sample size) of BPA concentration (ng/L) in natural surface waters (freshwater, estuaries and coasts) for each country. Individual BPA levels presented as grey points. (e) Percentage of plastic waste inadequately managed per country with high risk of polluting rivers and oceans (highlighted countries), and the mean plastic input from rivers into coastal outlets (presented as spatial points; t/year). Grey lines represent major river systems that lead to the coast. [†]Data obtained from Lebreton et al. (2017), and [‡]obtained from Jambeck et al. (2015) for the year 2010. Representative organism silhouettes obtained from phylopic (<http://phylopic.org/>)



ecosystems had the lowest mean BPA levels (Figure 1b). The four countries with the highest reported geometric mean BPA levels were Iran, Taiwan, Nigeria and Singapore (Figure 1c,d). East-Asia (China, Taiwan, Korea, Japan), North America (USA, Canada) and Europe were the most sampled regions. However, for many countries, there were no published BPA levels available, including Russia, many countries in Africa, the Middle East and South America. Both BPA levels and unmanaged plastic waste (Figure 1e) were most predominant in the tropic regions, where Asia accounted for 67% of the global input of marine plastic waste from river run-offs, which reflects both the areas of BPA manufacturing and the human population density (see Lebreton et al., 2017 for more detail). There was a positive relationship between the geometric mean BPA concentration and total plastic waste generated per year across different countries (Figure S5; $t_{43} = 3.3$, $R^2 = .2$, $p = .001$), while there was no relationship with the percentage of mismanaged waste ($t_{45} = -0.54$, $R^2 = .005$, $p = .6$).

3.2 | Overall effect of BPA exposure

The final data set comprised of 2,892 individual effect sizes from 174 studies on 88 different species. The full summary of observations for each response is provided in Table S6. Overall, exposure to BPA had a significantly negative effect on biological responses ($\ln R^2 = -0.19$; 95% CI $[-0.31, -0.07]$; $n = 2,892$ effect sizes; $p = .002$). Heterogeneity in the data set was high (I^2 total = 99%; Table S7) which is not surprising given the diverse species, number of traits and developmental stages included (Senior et al., 2016). Using meta-regression analyses including the five moderator variables (sex, biological levels, traits, taxonomic groups and developmental period) explained 23.33% (R^2) of the variance. Continuous moderators including BPA dosage, exposure period and test temperature explained 2.96% (R^2) of the total variance. BPA dosage had a significant effect on aquatic organisms, while the exposure period

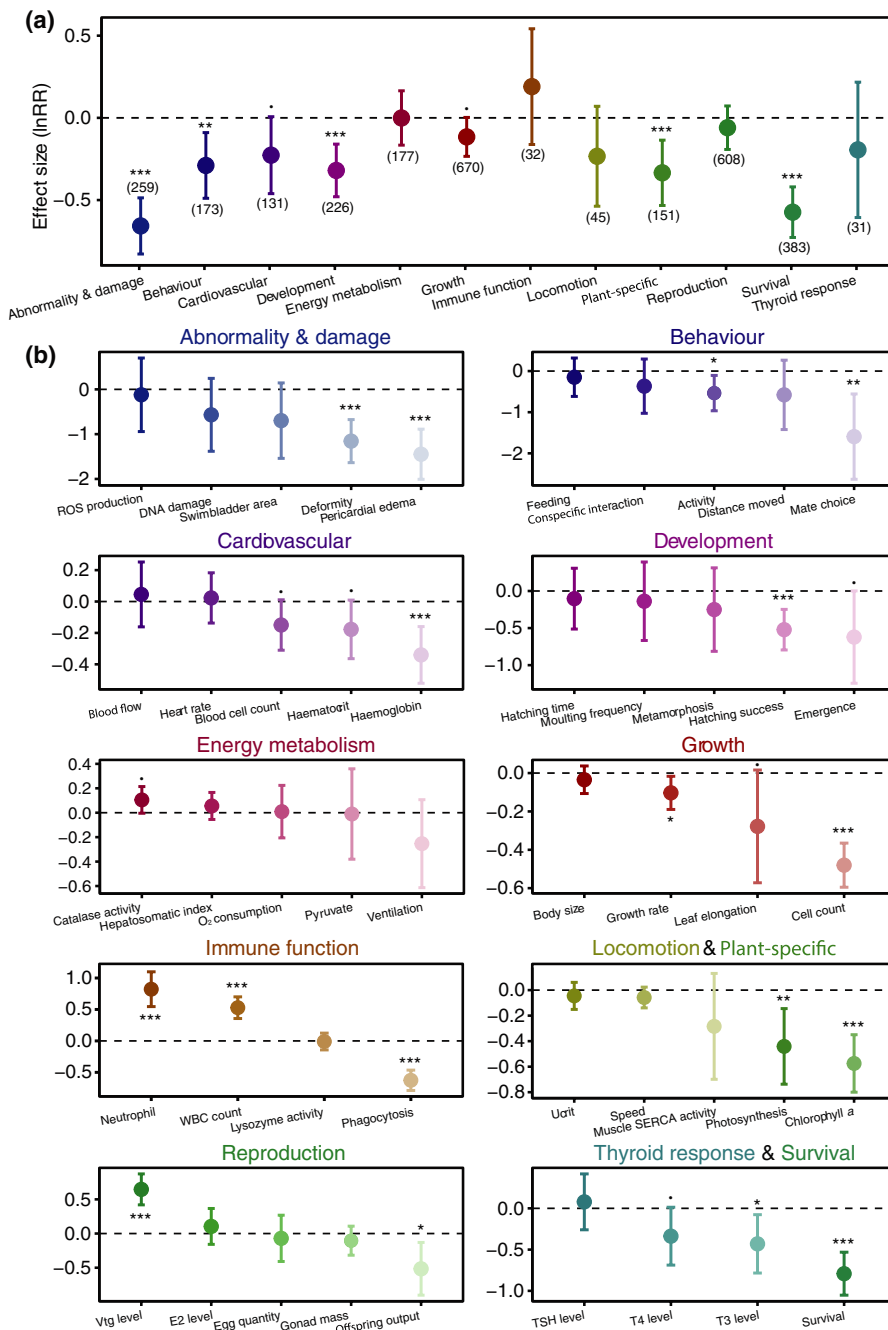


FIGURE 2 Effect of bisphenol A (BPA) exposure on different functional traits. (a) Overall effect sizes of BPA exposure on the 12 traits categories. (b) Effect sizes of BPA exposure of specific responses within each trait category. Each panel shows the five responses with the highest number of effect sizes (k). Data are presented as mean effect sizes (log response ratio; $\ln RR \pm 95\%$ CI, and numbers within parentheses represent the number of effect sizes in the sample. Horizontal dashed line at zero indicates no effect, and negative values represent negative effects on biological functions. Effect size were considered significant when 95% CI do not overlap with the zero line as denoted by asterisks (* $p < .05$, ** $p < .01$, *** $p < .001$). Sample size of specific responses provided in Table S6, and full summary of statistical outcome provided in Table S10 [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/gcb.15127)]

and the test temperature of exposure did not influence the effect of BPA on aquatic organisms (Table S8).

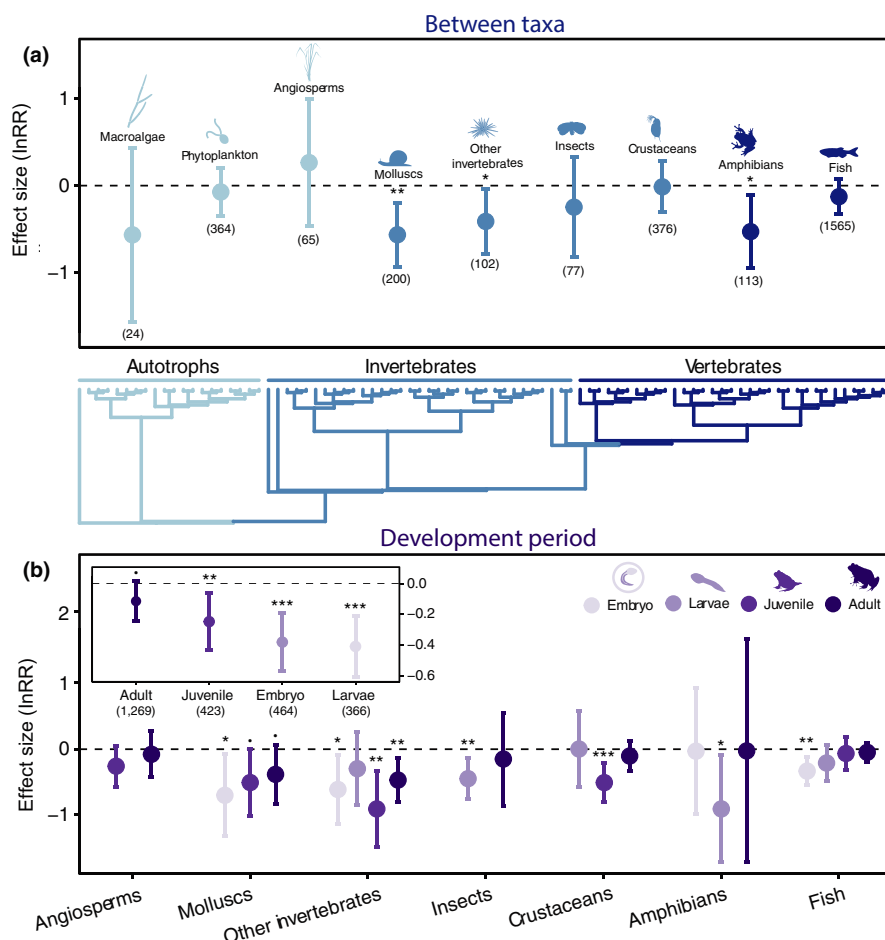
3.3 | Effect of BPA exposure across sex, biological level and functional traits

Studies that recorded sex (83 studies) showed that sex explained 12% of the variation in responses to BPA exposure. Across biological levels which explained 4.6% of the variation, cellular ($z = -2.23$; $n = 595$; $p = .026$), tissue ($z = -7.59$; $n = 75$; $p < .0001$), organ ($z = -3.4$; $n = 336$; $p = .0007$) and organism ($z = -4.35$; $n = 1,543$; $p < .0001$) level response were negatively affected by BPA exposure, while

macromolecular responses showed a marginally significant increase in response to BPA exposure ($z = 1.93$; $n = 343$; $p = .054$). All summary statistical outcomes for sex and hierarchical moderators are presented in Table S9.

We examined how BPA exposure affected 12 trait categories. We found that abnormality and damage ($z = -7.54$; $n = 259$; $p < .0001$), behaviour ($z = -2.84$; $n = 173$; $p = .005$), development ($z = -3.91$; $n = 226$; $p < .0001$), plant-specific traits ($z = -3.3$; $n = 151$; $p = .001$) and survival ($z = -7.3$; $n = 383$; $p < .0001$) were significantly reduced in the presence of BPA (Figure 2a). Cardiovascular ($z = -1.9$; $n = 131$; $p = .057$) and growth ($z = -1.9$; $n = 670$; $p = .055$) traits showed a two-tailed significant reduction from BPA exposure. Within each trait category, we performed individual analyses of up

FIGURE 3 Effect of bisphenol A (BPA) exposure on different taxa and developmental periods. (a) Effect size of BPA exposure across all nine taxonomic groups. The phylogenetic tree represents all 84 species in the study with their relative position in the tree. Blue colours represent three main biological domains (autotrophs, invertebrates, vertebrates). (b) Effect size of BPA exposure during development across taxa. Inset graph represents overall effect size for each development period. Purple shading represents development stages (embryo, larvae, juvenile, adult). For both graphs, data are presented as mean effect size (log response ratio (lnRR)) $\pm$ 95% CI, and numbers within parentheses are the number of effect sizes in the sample. Horizontal dashed line at zero indicates no effect, and negative direction represents a negative effect on the trait. Effect size were considered significant when 95% CI do not overlap with the zero line as denoted by asterisks (* $p < .1$, ** $p < .05$, *** $p < .01$, and **** $p < .001$). Representative organism silhouettes obtained from phylopic (<http://phylopic.org/>). Full summary of statistical results provided in Table S11 [Colour figure can be viewed at wileyonlinelibrary.com]



to five specific traits that were represented by the highest sample sizes (Figure 2b). Under the abnormality and damage category, morphological abnormalities ($z = -4.7$; $n = 85$; $p < .0001$) and pericardial oedema ($z = -5.07$; $n = 49$; $p < .0001$) increased the most with BPA exposure. Mate choice ($z = -3.01$; $n = 4$; $p = .003$) and general activity levels ($z = -2.44$; $n = 89$; $p = .01$) were significantly reduced by BPA exposure in the behavioural traits. Cardiovascular traits such as circulating haemoglobin levels ($z = -3.69$; $n = 13$; $p = .0002$) were significantly reduced, and in the development category, hatching success when exposed to BPA as embryos ($z = -3.73$; $n = 105$; $p = .0002$) were lower compared to controls. Within the energy metabolism traits, catalase activity as part of the antioxidant response ($z = 1.88$; $n = 106$; $p = .06$) showed a two-tailed significant increase from BPA exposure. Cell count (for unicellular organisms) and growth rate (for multicellular organisms) were significantly reduced by BPA in the growth category ($z = -8.18$; $n = 89$; $p < .0001$, and $z = -2.34$; $n = 381$; $p = .02$ respectively). Within the immune function category, both neutrophil ($z = 5.82$; $n = 9$; $p < .0001$) and white blood cell counts ($z = 6.01$; $n = 5$; $p < .0001$) significantly increased in response to BPA exposure, while phagocytosis ($z = -7.58$; $n = 7$; $p < .0001$) decreased from BPA exposure. Locomotion was not significantly affected by BPA exposure, while the plant-specific traits showed that both photosynthesis ($z = -2.9$; $n = 39$; $p < .0001$) and chlorophyll a levels ($z = -5$; $n = 112$; $p < .0001$) were significantly reduced by BPA

exposure. Within the thyroid responses category, triiodothyronine (T3) levels were significantly lower in the presence of BPA ($z = -2.39$; $n = 9$; $p = .02$), while thyroid-stimulating hormone (TSH) and thyroxine (T4) levels were not significantly affected ($z = 0.45$; $n = 7$; $p = .64$, and $z = -1.89$; $n = 15$; $p = .06$ respectively). Survival was significantly reduced by BPA exposure ($z = -5.94$; $n = 383$; $p < .0001$). All summary statistical outcomes for each response are presented in Table S10. To exclude potential taxonomic bias of the results (fish comprised of 54% of the effect sizes extracted), we ran the functional trait meta-analysis with the exclusion of fish and presented the results in Figure S6.

3.4 | Effect of BPA exposure between taxonomic groups and across development stage

The effect of BPA exposure varied between taxonomic groups, which explained 3.6% of the variation. Molluscs ($z = -3$; $n = 200$; $p = .003$), other invertebrates ($z = -2.4$; $n = 102$; $p = .03$) and amphibians ($z = -2.46$; $n = 113$; $p = .02$) were significantly affected by BPA exposure (Figure 3a). All developmental stages except for the adult stage were affected negatively by BPA exposure (embryo: $z = -4$; $n = 464$; $p < .0001$; larvae: $z = -4.04$; $n = 366$; $p < .0001$; juvenile: $z = -2.7$; $n = 423$; $p = .008$; Figure 3b), contributing to 1.9% of the variation. When sub-setting

developmental stage across taxonomic groups, there were taxon-specific responses. However, on average, younger individuals (either embryonic, larval or juvenile) of each taxon were more sensitive to BPA exposure than adults (Figure 3b). All statistical outcomes for taxon-specific and developmental response are provided in Table S11.

4 | DISCUSSION

We found that BPA was present in natural water bodies across the world, and on average, freshwater systems had higher BPA levels than marine and estuarine systems, and there was a strong correlation in BPA concentrations with the amount of plastic waste generated. However, there was considerable sample bias, and no BPA records were available from most of South America, and Africa. Areas with plastic waste generated (Jambeck et al., 2015; Lebreton et al., 2017) have potentially high BPA levels, and these should be of particular focus, especially when they coincide with areas of high biodiversity such as in South America and Africa. Even within countries for which records exist, it is unlikely that these represent a thorough coverage given the few available data points. Surface water concentrations can vary depending on the location, sampling time and season and analysis used (Flint et al., 2012; Lin et al., 2017; Ozhan & Kocaman, 2019). For example, surface water BPA levels can vary with seasonal rainfall and water temperature (Wu et al., 2013), so that aquatic organisms can be subjected to higher levels of BPA during certain seasons, and may be more sensitive to BPA at specific times when breeding or development coincides with periods of high BPA levels.

The scarcity of data on BPA levels in natural environments is particularly alarming because BPA has a negative effect on a broad range of biological functions. We showed that BPA exposure has an overall negative effect on aquatic organisms. The biological effects of BPA are most likely mediated by its endocrine-disrupting action, particularly of oestrogen receptors and thyroid hormone (Chapin et al., 2008; Kuiper et al., 1998). For example, within the reproductive category of our analysis, there was a significant increase in circulating vtg levels, and a significant decrease in reproductive output when exposed to BPA. Vitellogenin is a precursor protein for egg yolk development, and high vtg levels can cause feminization across many animal species (Levy, Lutz, Krüger, & Kloas, 2004; Lutz & Kloas, 1999; Matozzo, Gagné, Marin, Ricciardi, & Blaise, 2008; Sumpter & Jobling, 1995). Decreased reproductive output can alter population dynamics and persistence so that BPA has the potential to have serious consequences for biodiversity (Figure 4).

Bisphenol A inhibits thyroid hormone-mediated transcriptional activity by binding to thyroid hormone receptors (Gentilcore et al., 2013; Moriyama et al., 2002). Thyroid hormones are important for regulating a myriad of physiological processes including development, metabolism and muscle function (Mullur, Liu, & Brent, 2014), with downstream effects on locomotor performance and activity levels, for example (Casals-Casas & Desvergne, 2011). Our meta-analysis shows limited support for disruption of energy balance,

locomotion and muscle function. However, these traits are less studied when compared to reproductive and developmental traits. Future studies should examine if the reduction of thyroid function from BPA exposure causes disruption in whole organism energy balance (Ortiz-Villanueva et al., 2017; Sadoul et al., 2017) and other ecologically relevant responses such as activity, behaviour, growth and development. Nevertheless, there is clear evidence of disruption of biological systems by BPA exposure, as many of the traits affected are directly related to fitness (e.g. growth, survival, reproduction). We also observed that the effects of BPA did not significantly differ between sexes, but showed a significant effect for studies that combined both males and females. The significance may be an artefact of the differences in biological traits measured between studies that stated sex and those that did not.

There is mixed evidence in the literature that BPA accumulates in organisms (Chen, Gwo, Wang, & Chen, 2014; Corrales et al., 2015; Staples et al., 1998). In laboratory conditions, BPA is metabolized rapidly (<6 hr) in fish bile (Huang et al., 2018; Lindholm, Pedersen, & Pedersen, 2000; Lindholm, Wynne, Marriott, Pedersen, & Bjerregaard, 2003), and the low to moderate hydrophobicity of BPA would indicate a modest capacity for bioaccumulation (Cousins et al., 2002; Heinonen, Honkanen, Kukkonen, & Holopainen, 2002). However, various field studies showed that BPA can accumulate in fish (Belfroid, Velzen, Horst, & Vethaak, 2002; Lee et al., 2015), indicating that ecological interactions outside of laboratory conditions play a role in how BPA is accumulated, metabolized and excreted (Figure 4). Various abiotic and biotic factors can affect the rate of accumulation and metabolism of BPA, including taxon-specific responses in metabolic pathways (Kang, Katayama, & Kondo, 2006; Xiao, Wang, Zhou, & Huang, 2019), body size and ontogeny (Peng et al., 2018), the stress levels of the organisms (Koponen & Kukkonen, 2002), temperature (Heinonen et al., 2002; Honkanen & Kukkonen, 2006) and the interactions with other natural and synthetic chemicals (Borirukwitsak, Keenan, & Gauchotte-Lindsay, 2012; Chen et al., 2017). These factors may explain differences between laboratory and field studies. We show that molluscs, other invertebrates and amphibians are among the most susceptible taxa to BPA exposure. Additionally, there is a clear trend that embryos, larvae and juveniles are the most sensitive life-history stages across all animal groups. This trend has broader ramifications for aquatic food webs, as larvae are important prey and consumers in aquatic systems (Gove et al., 2019; Llopiz & Cowen, 2009; Stier, Hein, Parravicini, & Kulbicki, 2014), and increased larval mortality can impact population dynamics. However, more studies on underrepresented taxa are required to draw generalized conclusions about the developmental effects of BPA.

An important challenge for understanding how organisms respond to endocrine-disrupting compounds is to determine interactions with other stressors including other abiotic factors such as temperature and acidification, and biotic interactions resulting from density changes and habitat alteration, for example (DeCourten, Romney, & Brander, 2019). To date, only a handful of

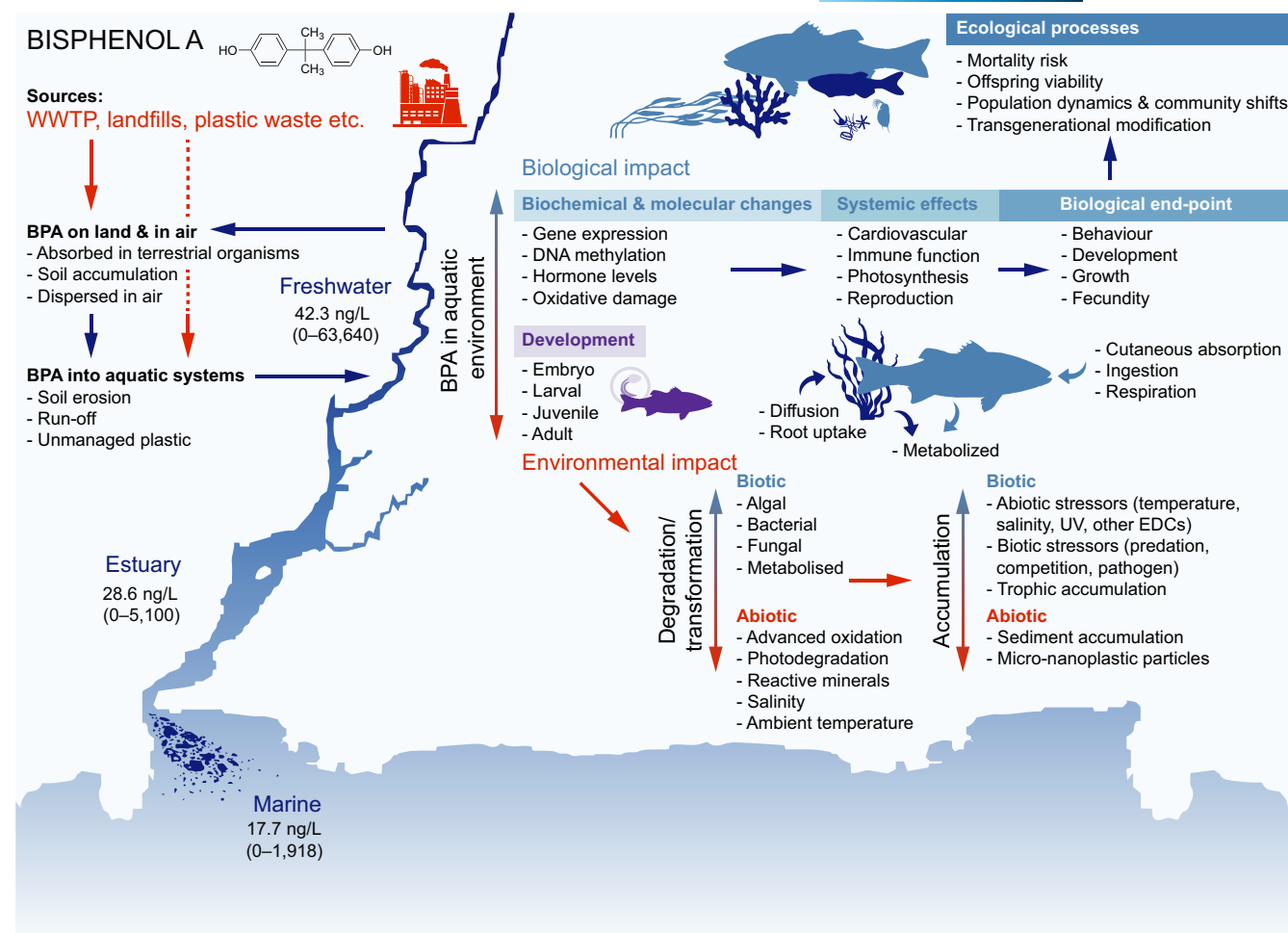


FIGURE 4 Schematic outline of the source, dynamic effect and fate of bisphenol A (BPA) in aquatic environments. Biological impact shows the cascade effects on aquatic organisms from the molecular/cellular responses to impacts on ecological processes. The schematic of the biological impact shown is biased towards fish, because fish represent the most studied animal group. Environmental impact shows how BPA is either degraded, transformed or accumulated through abiotic and biotic processes. Degradation/transformation represents known abiotic and biotic factors that breakdown or convert BPA to other chemicals. Accumulation represents abiotic and biotic factors that influence the accumulation potential or rate of accumulation of BPA in the biota or the environment. Numbers for each aquatic system represents the geometric mean surface water BPA levels from the meta-analysis. The minimum and maximum detected concentrations are presented in brackets. EDC, endocrine-disrupting chemical; UV, ultraviolet radiation; WWTP, wastewater treatment plant. Aquatic organism silhouettes obtained from phylopic (<http://phylopic.org/>) [Colour figure can be viewed at wileyonlinelibrary.com]

studies have examined the interaction of BPA with other environmental stressors. Our analysis shows that BPA has far-reaching biological effects, but these may be (or may have been) modulated by other external stressors that alter the sensitivity of organisms to BPA or modulate its physiological effects. Temperature (Heinonen et al., 2002; Little & Seebacher, 2015), hypoxia (Cypher, Ickes, & Bagatto, 2015) and UV-B exposure (Koponen & Kukkonen, 2002; Koponen, Tuikka, & Kukkonen, 2007) can further exacerbate the effects of BPA, or the presence of BPA enhances the physiological disruption from these stressors. These interactions need to be clarified experimentally across taxa. A common issue facing the ecotoxicology research is that variations in environmental conditions are not accounted for in environmental risk assessments, which may cause a mismatch between laboratory testing and responses in natural environments (DeCourten et al., 2019).

Examining multiple stressors can provide more ecological relevant response for informing conservation management (Todgham & Stillman, 2013).

5 | CONCLUSION

The volume of plastic waste entering oceans is huge and increasing (Wang, He, & Sen, 2019), and it is predicted that by 2025, 150 million metric tons of plastic waste will enter oceans globally (Jambeck et al., 2015). Aquatic ecosystems will therefore be subjected to greater concentrations of EDC, which will represent major concerns for environmental health. Overall, our findings show that BPA negatively affects aquatic organisms, and that early development periods across all taxa are most sensitive to BPA, which can ultimately affect

population dynamics. This study synthesizes the current literature of the effects of BPA, as well as the current BPA levels in the environment and in aquatic organisms, providing important ecological implications for the conservation and management of aquatic ecosystems, as well as elucidating critical research gaps needed to fully comprehend the effects of BPA.

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CONFLICT OF INTEREST

The authors declare no competing interests.

AUTHOR CONTRIBUTION

N.C.W. collected and analysed the data, produced the figures, wrote the initial manuscript. N.C.W. and F.S. conceived the study, interpreted the data, drafting and finalizing the paper.

DATA AVAILABILITY STATEMENT

All data sets generated and analysed during the study are available on the GitHub repository: <https://github.com/nicholaswunz/BPA-meta-analysis>.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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